

# Spatial Graph Signal Interpolation with an application for Merging BCI Datasets with various Dimensionalities

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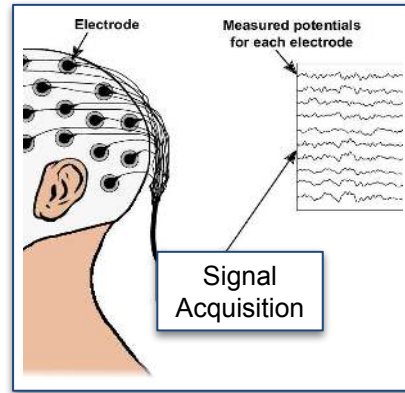
## ICASSP 2023



Région  
BRETAGNE

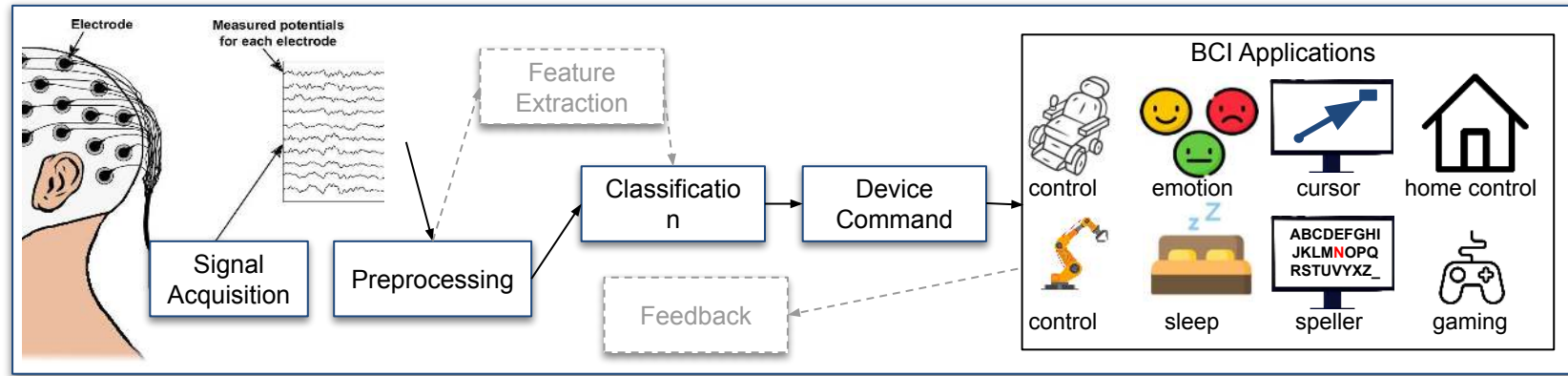
08/06/2023

# Context - Brain Computer Interface (BCI)

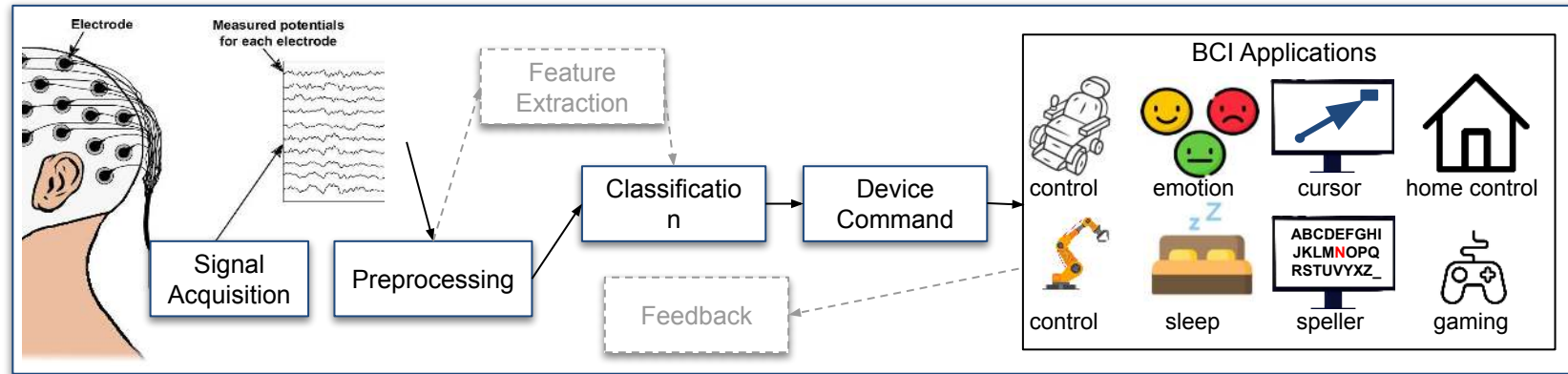


ElectroEncephaloGraphy  
(EEG)

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## Challenges:

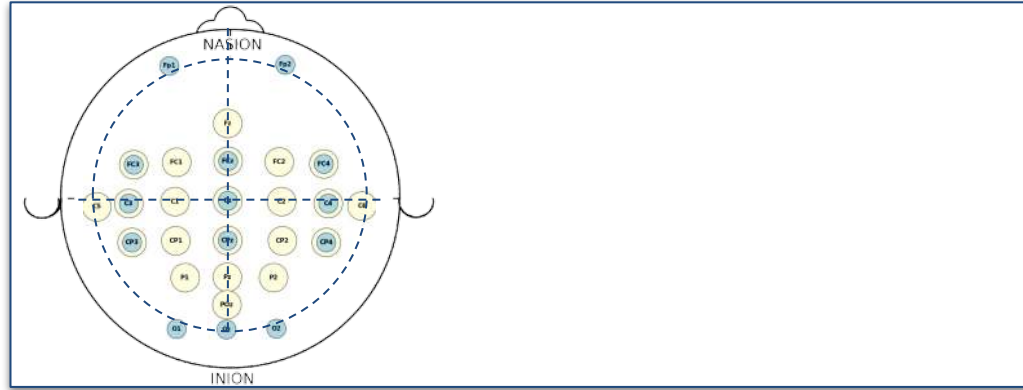
- Lack of big datasets.
- Small dataset with different recording setups.

Electrodes layout, sampling frequency, filters, ...



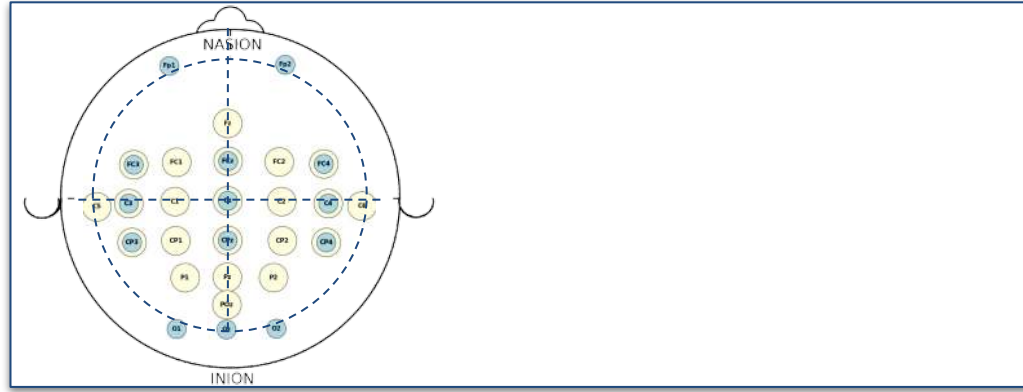
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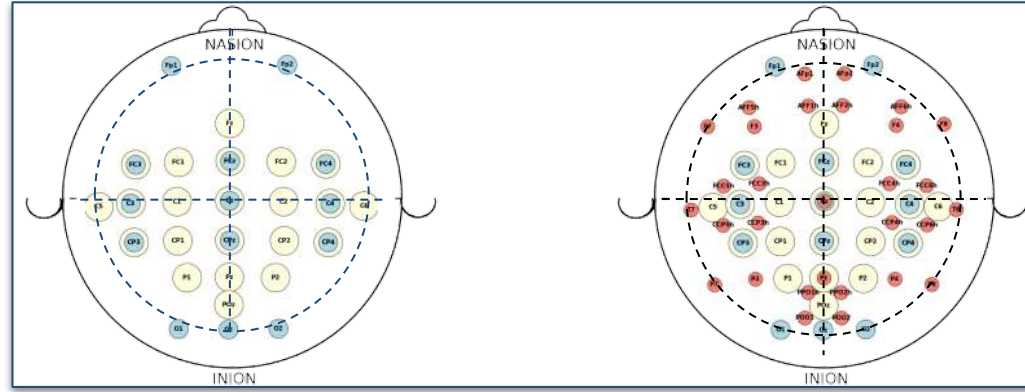


Few approaches have been proposed to unify the spatial aspect of the EEG setups.

- Reduce the spatial dimension:
  - Keeping the intersection
  - Dimension reduction methods (ex : PCA)

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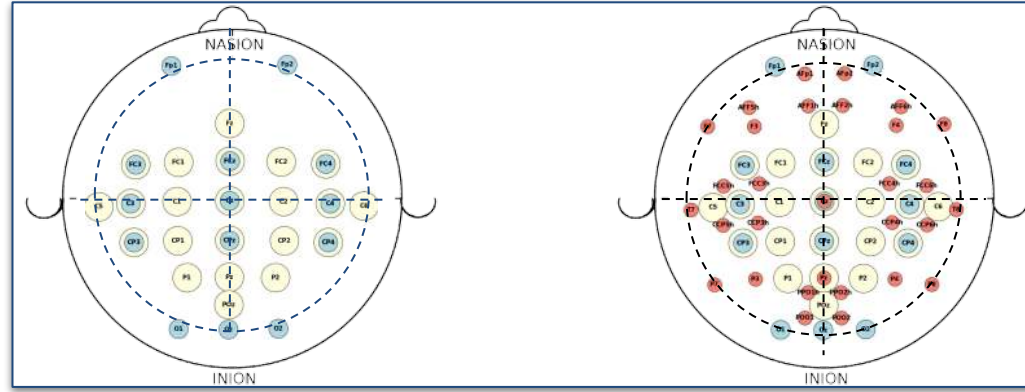


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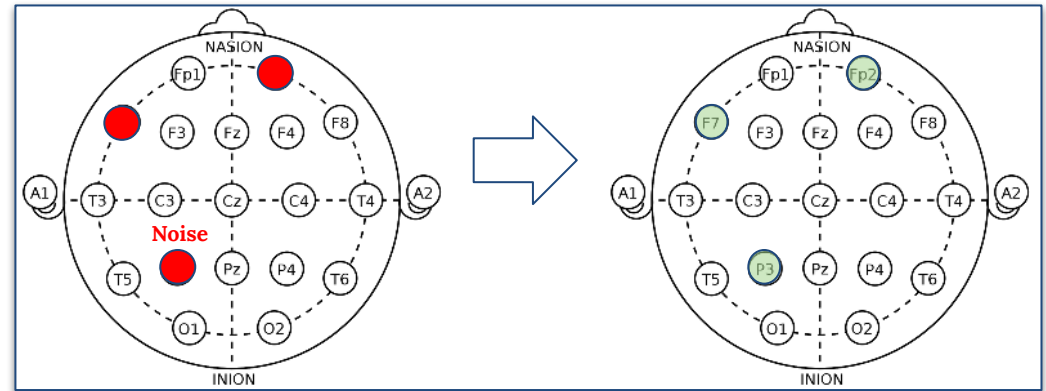
No information loss



# Context - Electrodes / Graph interpolation

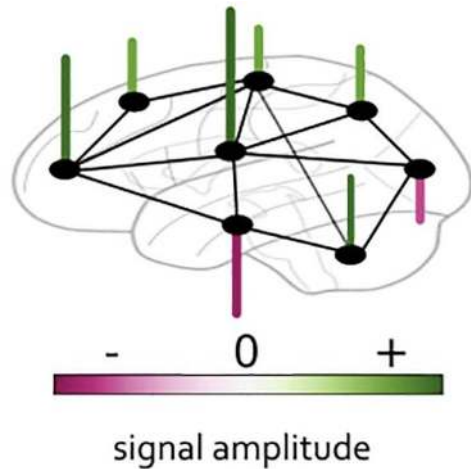
Interpolating electrodes has been mainly used to recover signal from noisy electrodes

Mainly using is Spherical Spline [1]



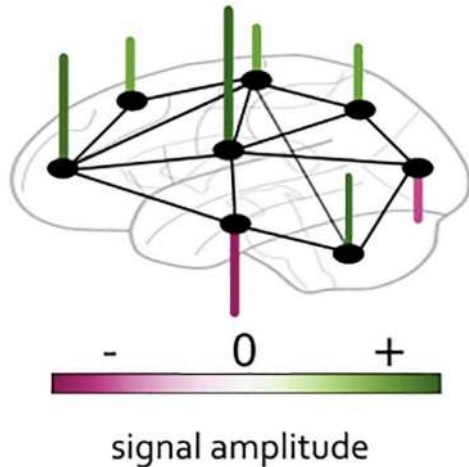
# Scientific challenges

**Our proposition: Interpolating spatial EEG using Graph Signal Processing**



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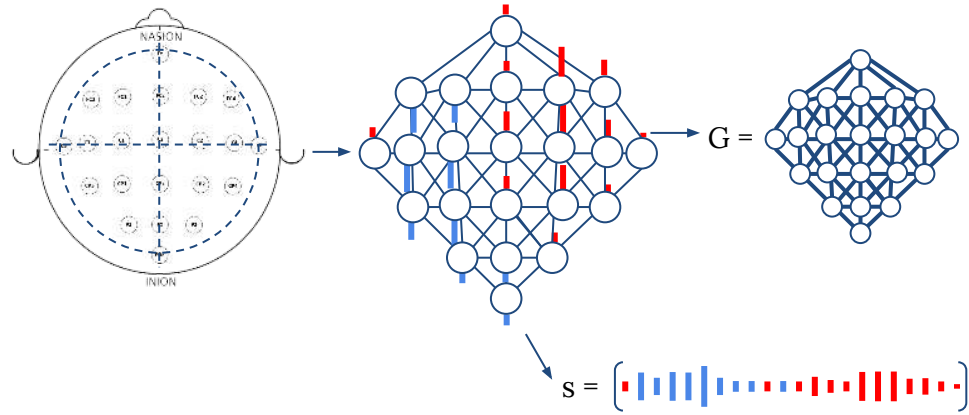


Research questions:

- 1) How to use GSP to interpolate electrodes?
  - Which graph?
  - Which interpolation criterion
- 2) Does unifying multiple EEG datasets with interpolation improve brain decoding?

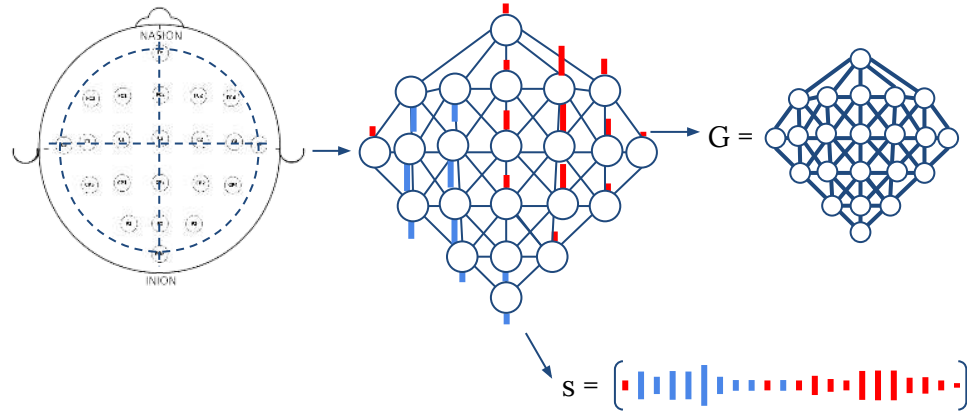
# Methods

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With  $D$  and  $W$ , the Degree and weights matrix of  $G$ , we have

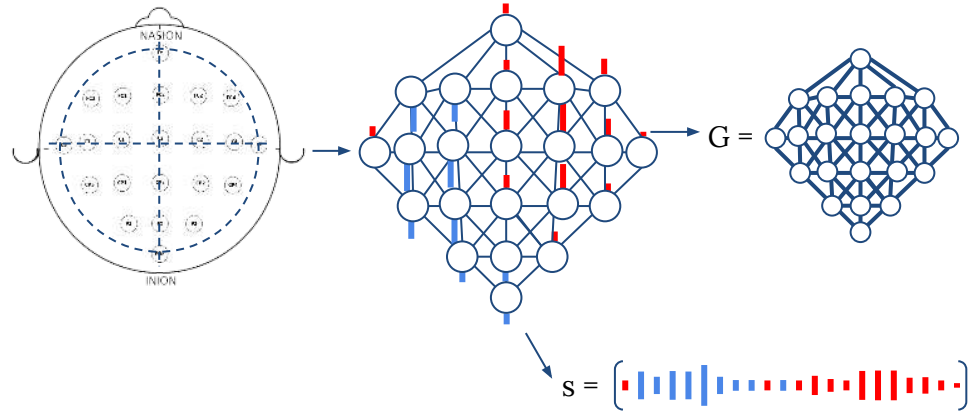
$\mathbf{L} = \mathbf{D} - \mathbf{W}$  , with  $\mathbf{L}$  the Laplacian

Smoothness of the signal over the graph is defined by:

$$\sigma(\mathbf{s}) = \mathbf{s}^\top \mathbf{L} \mathbf{s} = \sum_{i=1}^{|\mathcal{V}|} \sum_{j=1}^{|\mathcal{V}|} W_{ij} (s_i - s_j)^2$$

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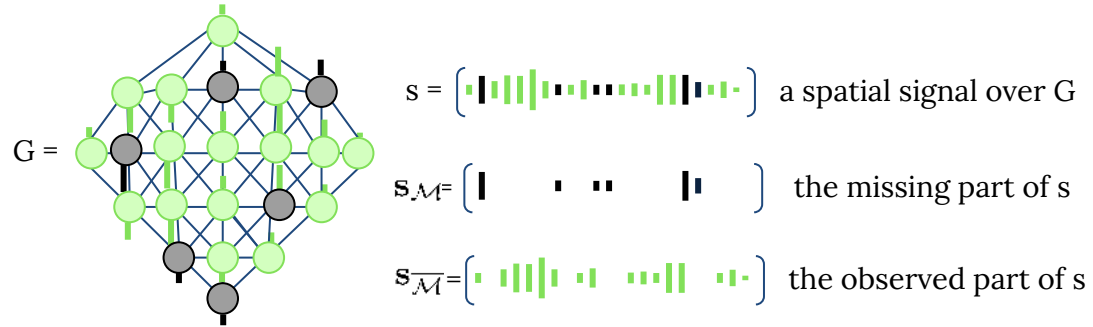
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Interpolation problem formulation:



- 1) How to find SM? Which interpolation criterion?
- 2) How to build a Graph?

## 1 How to find $\mathbf{s}_{\mathcal{M}}$ ?

Interpolation criterion : **smoothness**

Minimizing  $\sigma(\mathbf{s}) = \mathbf{s}^\top \mathbf{L} \mathbf{s} = \sum_{i=1}^{|\mathcal{V}|} \sum_{j=1}^{|\mathcal{V}|} W_{ij} (s_i - s_j)^2$

We found a closed form of  $\sigma(\mathbf{s})$  that provides the optimal  $\mathbf{s}_{\mathcal{M}}$

$$(A) \quad \mathbf{s}_{\mathcal{M}} = -\mathbf{L}_{\mathcal{M}}^{-1} \mathbf{L}_{\mathcal{M}\overline{\mathcal{M}}} \mathbf{s}_{\overline{\mathcal{M}}}$$

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## 2 How to build G?

We learn G, from real data using gradient descent.

- 1) Initialize a connected graph G
- 2) Create virtual reconstruction problems
- 3) Reconstruct the signal using (A)
- 4) Update G based on the error of reconstruction
- 5) Repeat 2) to 4) until the error is low

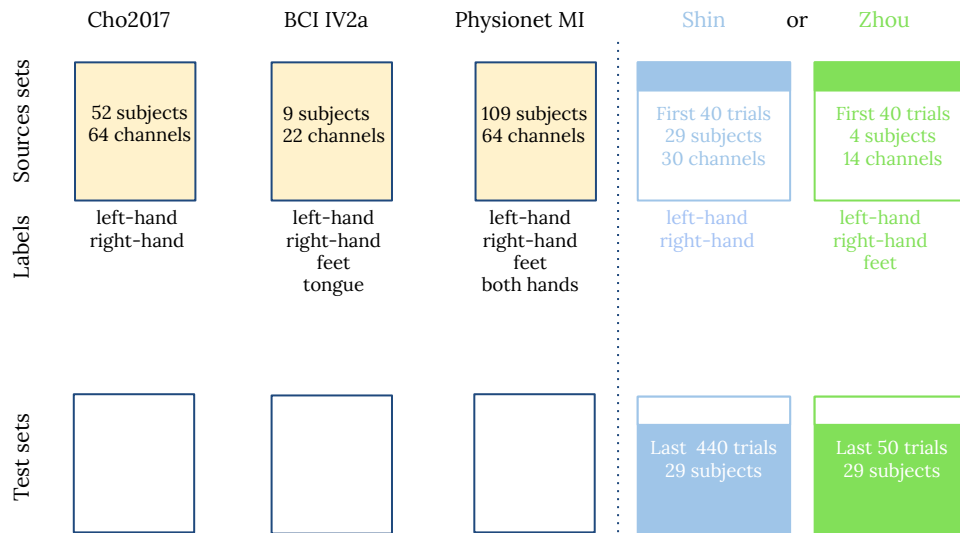


# Experiments

## What is the added value brought by interpolated data?

Does unifying multiple EEG datasets with interpolation improve brain decoding?

We experiment brain decoding on the following realistic setup:



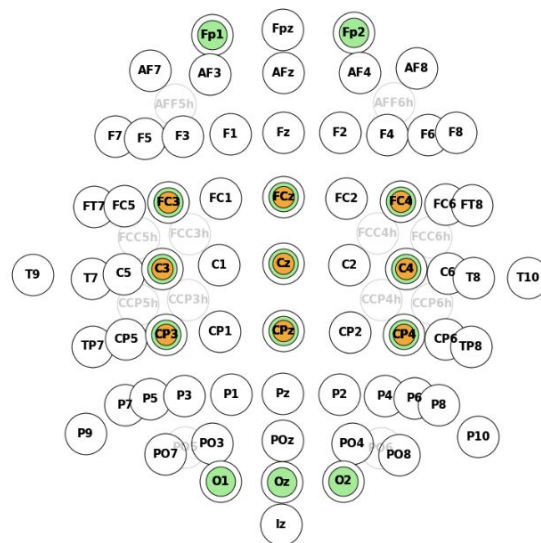
# Results

## Task : Brain decoding

|              | Cho2017                 | BCI IV2a                                  | Physionet MI                                  | Shin                           | or | Zhou                            |
|--------------|-------------------------|-------------------------------------------|-----------------------------------------------|--------------------------------|----|---------------------------------|
| Sources sets | 52 subjects             | 9 subjects                                | 109 subjects                                  | First 40 trials<br>29 subjects |    | First 40 trials<br>4 subjects   |
| Labels       | left-hand<br>right-hand | left-hand<br>right-hand<br>feet<br>tongue | left-hand<br>right-hand<br>feet<br>both hands | left-hand<br>right-hand        |    | left-hand<br>right-hand<br>feet |
| Test sets    |                         |                                           |                                               | Last 440 trials<br>29 subjects |    | Last 50 trials<br>29 subjects   |

## Classification accuracy

|         | Zhou              |          |
|---------|-------------------|----------|
|         | Acc               | N        |
| $\cap$  | <u>61.2 ± 2.0</u> | <u>9</u> |
| Dataset | 56.2 ± 4.8        | 14       |
| $\cup$  | 46.4 ± 2.8        | 66       |



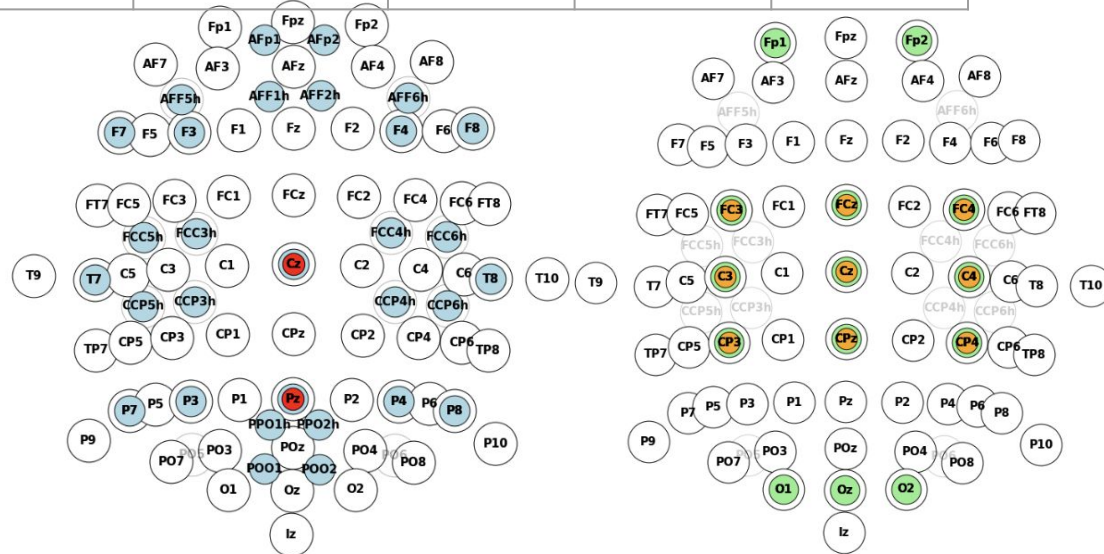
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## Classification accuracy

|         | Shin           |    | Zhou           |    |
|---------|----------------|----|----------------|----|
|         | Acc            | N  | Acc            | N  |
| $\cap$  | $53.2 \pm 2.8$ | 2  | $61.2 \pm 2.0$ | 9  |
| Dataset | $63.2 \pm 2.3$ | 22 | $56.2 \pm 4.8$ | 14 |
| $\cup$  | $62.3 \pm 2.1$ | 76 | $46.4 \pm 2.8$ | 66 |



# Conclusions

- New and efficient electrode interpolation technique exploiting GSP tools
- Illustrated the interest of our method to homogenize datasets
  - Our code is open
  - Many more details in our paper!

